

Program : Diploma in Architecture	
Course Code : 2182	Course Title: Fundamentals of Architecture
Semester : 2	Credits: 3
Course Category: Engineering Science	
Periods per week: 3 (L:2, T:1, P:0)	Periods per semester: 45

Course Objectives:

- To enable the students, identify the elements and principles of architecture applied in building.
- To comprehend the concepts and architectural features of buildings designed by well-known Indian and world architects.

Course Prerequisites:

Topic	Course code	Course name	Semester
Fundamentals of 3D objects		Engg: graphics	1

Course Outcomes

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive level
CO1	Explain the elements and principles of design	10	Understanding
CO2	Interpret architectural space and outline the design process	11	Understanding
CO3	Explain the design concept and architectural features of buildings in world architecture	11	Understanding
CO4	Explain the design concept and architectural features of buildings in Indian architecture	11	Understanding
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						2
CO3	3					2	
CO4	3					2	

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain the elements and principles of design		
M1.01	Summarize elements of design	4	Understanding
M1.02	Explain basic principles of Design	4	Understanding
M1.03	Explain creative principles of design	2	Understanding
Contents: Primary elements of design - point, line, plane and volume - characteristics and uses, Form- Definition, visual and relational properties, regular and irregular forms. Colour - Definition, Qualities - hue, value and intensity - tint and shade - Colour scheme (spectrum) - primary, secondary and tertiary colours - colour wheel - warm, cool, dark, light colours - complementary colours, analogues colour, monochromatic colour, triadic colours etc.- Texture- Definition and explanation of rough and smooth textures. The basic principles of composition such as Balance, Proportion, Scale, Rhythm, Contrast, Harmony and Character - Brief explanation with examples of buildings. Creative principles of design - Function, Strength and Aesthetics or Appearance			
CO2	Interpret architectural space and outline the design process		
M2.01	Explain space in architecture	2	Understanding
M2.02	Explain spatial organizations	6	Understanding
M2.03	Explain architectural design process	3	Understanding
	Series Test-I	1	
Contents: Definition of space, Explanation of interior and exterior spaces; Characteristics of interior space- Quantity and Quality; Spatial organizations - Centralized, Linear, Radial, and Clustered and Grid organizations - Brief explanation only. The steps involved in architectural design process - program analysis, preliminary studies, intermediate studies, presentation drawings - brief explanation only			

CO3	Explain the design concept and architectural features of buildings in world architecture		
M3.01	Identify the design concept and architectural features of given world - famous structures	5	Understanding
M3.02	Illustrate architectural features of given world-famous structures	6	Applying
Contents: Design concept, architectural features along with a sketch of the following famous structures - Gustave Eiffel - Eiffel Tower FL Wright - Falling Water, Pennsylvania, USA Le Corbusier - Villa Savoye, France Mies Van der Rohe - Farnsworth House, USA Antonio Gaudi - Casa Batllo, Barcelona, Spain Jorn Utzon - Sydney Opera House I M Pei - Louvre pyramid, France Santiago Calatrava - HSB Turning torso, Sweden Geoffrey Bawa - Sri - Lankan Parliament Complex Zaha Hadid - Heydar Aliyev Center Azerbaijan. (A portfolio can be prepared at the end of the module)			
CO4	Explain the design concept and architectural features of buildings in Indian architecture		
M4.01	Identify the design concept and architectural features of below listed buildings in Indian architecture	6	Understanding
M4.02	Illustrate below listed buildings in Indian architecture	5	Applying
	Series Test - I	1	
Contents: Design concept and overall features with sketch view of the following famous structures. Edwin Lutyens - Rashtrapathi Bhavan, New Delhi Louis I Kahn - IIM, Ahmadabad Charles Correa - Kovalam Beach Resort, Tube house B.V Doshi - Sangath, at Ahmadabad Hafees Contractor - Buckley Court apartment, Colaba Achyut P Kanvinde - IIT Kanpur Laurie Baker - CDS, Trivandrum Raj Rewal - Permanent Exhibition Complex, Pragathi maidan, New Delhi (A portfolio can be prepared at the end of the module)			

Text / Reference:

T/R	Book Title/Author
T1	Francis DK Ching - Architecture Form, Space and Order
R2	Sabine Thiel Siling - Icons of Architecture - The twentieth century
R3	Ahmed A Kasu - Interior Design
R4	Banister Fletcher – History of world Architecture

Online resources:

Sl. No	Website Link
1	https://www.slideshare.net/